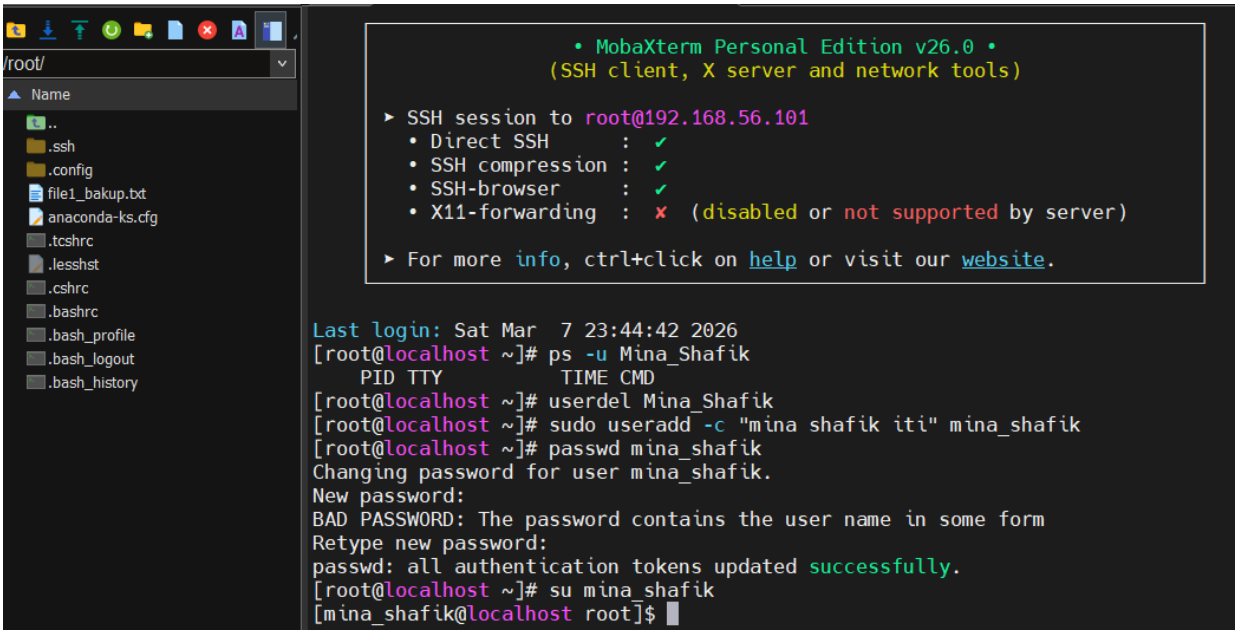
1. Create a user account with the following attribute

Username: yourname

Fullname/comment: yourname iti

Password: yourname

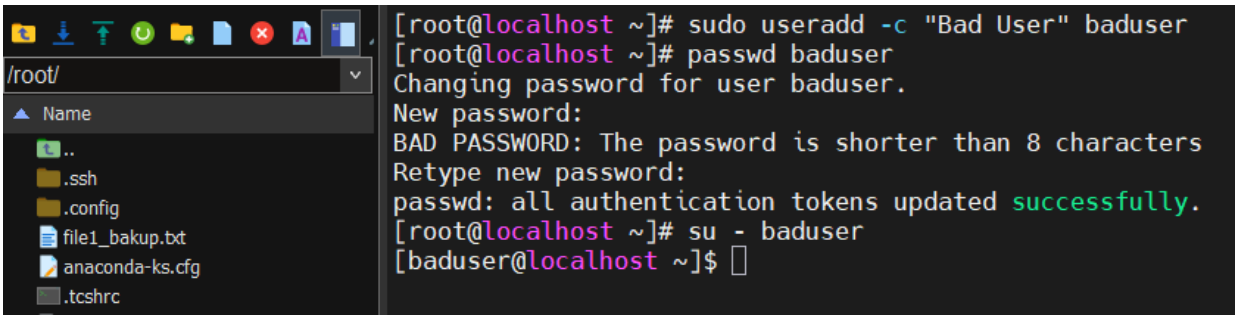


2. Create a user account with the following attribute

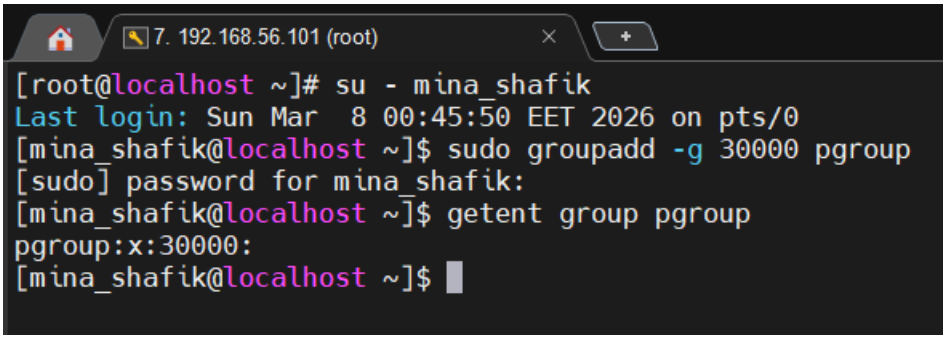
Username: baduser

Full name/comment: Bad User

Password: baduser



3. Create a supplementary (Secondary) group called pgroup with group ID of 30000



4. Create a supplementary group called badgroup

```
7. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ sudo groupadd badgroup
[mina_shafik@localhost ~]$ getent group badgroup
badgroup:x:30001:
[mina_shafik@localhost ~]$
```

5. Add yourname user to the pgroup group as a supplementary group

```
7. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ sudo usermod -aG pgroup mina_shafik
[mina_shafik@localhost ~]$ groups mina_shafik
mina_shafik : mina_shafik root wheel pgroup
[mina_shafik@localhost ~]$
```

6. Modify the password of yourname's account to password

```
7. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ sudo passwd mina_shafik
Changing password for user mina_shafik.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[mina_shafik@localhost ~]$
```

7. Modify yourname's account so the password expires after 30 days

```
7. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ sudo chage -M 30 mina_shafik
[mina_shafik@localhost ~]$ sudo chage -l mina_shafik
Last password change                                    : Mar 07, 2026
Password expires                                        : Apr 06, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 30
Number of days of warning before password expires       : 7
[mina_shafik@localhost ~]$
```

8. Lock bad user account so he can't log in

```
7. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ sudo usermod -L baduser
[mina_shafik@localhost ~]$ su - baduser
Password:
su: Authentication failure
[mina_shafik@localhost ~]$
```

9. Delete bad user account

```
7. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ sudo userdel baduser
[mina_shafik@localhost ~]$ sudo userdel -r  baduser
userdel: user 'baduser' does not exist
[mina_shafik@localhost ~]$ getent passwd baduser
[mina_shafik@localhost ~]$
```

10. Delete the supplementary group called badgroup.

```
7. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ sudo groupdel badgroup
[mina_shafik@localhost ~]$ getent group badgroup
[mina_shafik@localhost ~]$ groups mina_shafik
mina_shafik : mina_shafik root wheel pgroup
[mina_shafik@localhost ~]$
```

11. Create a folder called myteam in your home directory and change its permissions to read only for the owner.

```
2. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ mkdir ~/myteam
[mina_shafik@localhost ~]$ chmod 400 ~/myteam
[mina_shafik@localhost ~]$ ls -ld ~/myteam
dr--------. 2 mina_shafik mina_shafik 6 Mar  8 11:58 /home/mina_shafik/myteam
[mina_shafik@localhost ~]$
```

12. Log out and log in by another user

```
2. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ su - user1
Password:
Last login: Sun Mar  8 12:06:23 EET 2026 on pts/0
[user1@localhost ~]$
```

13. Try to access (by cd command) the folder (myteam)

```
2. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ su - user1
Password:
Last login: Sun Mar  8 12:06:23 EET 2026 on pts/0
[user1@localhost ~]$ cd ~/myteam
-bash: cd: /home/user1/myteam: No such file or directory
[user1@localhost ~]$ ls -d ~/myteam
ls: cannot access '/home/user1/myteam': No such file or directory
[user1@localhost ~]$
```

14. Using the command Line: Change the permissions of oldpasswd file to give owner read and write permissions and for group write and execute and execute only for the others (using chmod in 2 different ways)

```
3. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ chmod 631 oldpasswd
[mina_shafik@localhost ~]$ chmod u=rw,g=wx,o=x oldpasswd
[mina_shafik@localhost ~]$ ls -l oldpasswd
-rw--wx--x. 1 mina_shafik mina_shafik 0 Mar  8 12:53 oldpasswd
[mina_shafik@localhost ~]$
```

15. Change your default permissions to be no permission to everyone then create a

directory and a file to verify.

```
3. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ umask 777
[mina_shafik@localhost ~]$ mkdir testdir
[mina_shafik@localhost ~]$ ls
myteam  oldpasswd  testdir
[mina_shafik@localhost ~]$ touch testfile
[mina_shafik@localhost ~]$ ls
myteam  oldpasswd  testdir  testfile
[mina_shafik@localhost ~]$ ls -l
total 0
dr--------. 2 mina_shafik mina_shafik 6 Mar  8 11:58 myteam
-rw--wx--x. 1 mina_shafik mina_shafik 0 Mar  8 12:53 oldpasswd
d---------. 2 mina_shafik mina_shafik 6 Mar  8 13:06 testdir
----------. 1 mina_shafik mina_shafik 0 Mar  8 13:06 testfile
[mina_shafik@localhost ~]$
```

16. Create a file with permission 444. Try to edit in it and to remove it? Note what happened

```
3. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ touch myfile
[mina_shafik@localhost ~]$ ls
myfile  myteam  oldpasswd  testdir  testfile
[mina_shafik@localhost ~]$ chmod 444 myfile
[mina_shafik@localhost ~]$ ls -l myfile
-r--r--r--. 1 mina_shafik mina_shafik 0 Mar  8 13:30 myfile
[mina_shafik@localhost ~]$ vi myfile
[mina_shafik@localhost ~]$
```

17. What is the difference between the “x” permission for a file and for a directory?

For a file, x means the file can be executed as a program or script.

For a directory, x means you can enter/traverse the directory.

```
Last login: Sun Mar  8 12:47:27 2026 from 192.168.56.1
[root@localhost ~]# su - mina_shafik
Last login: Sun Mar  8 12:47:36 EET 2026 on pts/0
[mina_shafik@localhost ~]$ ls
mydir  mydtxt.txt  myfile  myteam  oldpasswd  testdir  testfile
[mina_shafik@localhost ~]$ ls -ld mydtxt.txt
-r--r--r--. 1 mina_shafik mina_shafik 0 Mar  8 14:14 mydtxt.txt
[mina_shafik@localhost ~]$ chmod +x mydtxt.txt
[mina_shafik@localhost ~]$ ls -ld mydtxt.txt
-r-xr-xr-x. 1 mina_shafik mina_shafik 0 Mar  8 14:14 mydtxt.txt
[mina_shafik@localhost ~]$ chmod 444 mydir
[mina_shafik@localhost ~]$ chmod +x mydir
[mina_shafik@localhost ~]$ ls -ld mydir
dr-xr-xr-x. 2 mina_shafik mina_shafik 6 Mar  8 13:40 mydir
[mina_shafik@localhost ~]$
```

18. What are the minimum permission needed for:

<u>Copy a directory (permission for source directory and permissions for target parent directory)</u>

Source directory:

Need read (r) to list files

Need execute (x) to access/traverse the directory and its contents

Target parent directory:

Need write (w) to create new directory in it, Need execute (x) to access the directory

```
4. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ mkdir ~/source_dir
[mina_shafik@localhost ~]$ touch ~/source_dir/file1
[mina_shafik@localhost ~]$ mkdir ~/target_dir
[mina_shafik@localhost ~]$ chmod 500 ~/source_dir/
[mina_shafik@localhost ~]$ chmod 700  ~/target_dir/
[mina_shafik@localhost ~]$ cp -r ~/source_dir/ ~/target_dir/
[mina_shafik@localhost ~]$ ls  ~/target_dir/
source_dir
[mina_shafik@localhost ~]$ ls -ld  ~/target_dir/
drwx------. 3 mina_shafik mina_shafik 24 Mar  8 14:37 /home/mina_shafik/target_dir/
[mina_shafik@localhost ~]$
```

Copy a file (permission for source file and and permission for target parent directory)

Source file:

Need read (r)to read the content

Target parent directory:

Need write (w) to create the new file

Need execute (x) to access the directory

```
[mina_shafik@localhost ~]$ touch ~/source_file
[mina_shafik@localhost ~]$ ls
mydir  mydtxt.txt  myfile  myteam  oldpasswd  source_dir  source_file  target_dir  testdir  testfile
[mina_shafik@localhost ~]$ chmod 400 ~/source_file
[mina_shafik@localhost ~]$ chmod 700 ~/target_dir/
[mina_shafik@localhost ~]$ cp ~/source_file ~/target_parent/
cp: cannot create regular file '/home/mina_shafik/target_parent/': Not a directory
[mina_shafik@localhost ~]$ cp ~/source_file ~/target_dir
[mina_shafik@localhost ~]$ ls -ld ~/target_dir/
drwx------. 3 mina_shafik mina_shafik 43 Mar  8 14:48 /home/mina_shafik/target_dir/
[mina_shafik@localhost ~]$
```

Delete a file

File itself:

No specific permission required on the file

Parent directory:

Need write (w) to remove an entry

Need execute (x) to access the directory

```
[mina_shafik@localhost ~]$ touch ~/target_dir/file_to_delete
[mina_shafik@localhost ~]$ chmod 400 ~/target_dir/file_to_delete
[mina_shafik@localhost ~]$ chmod 700 ~/target_dir
[mina_shafik@localhost ~]$ rm ~/target_dir/file_to_delete
rm: remove write-protected regular empty file '/home/mina_shafik/target_dir/file_to_delete'? y
[mina_shafik@localhost ~]$ ls
mydir  mydtxt.txt  myfile  myteam  oldpasswd  source_dir  source_file  target_dir  testdir  testfile
[mina_shafik@localhost ~]$ ls -ld ~/target_dir/
drwx------. 3 mina_shafik mina_shafik 43 Mar  8 14:54 /home/mina_shafik/target_dir/
[mina_shafik@localhost ~]$ cd ~/target_dir/
[mina_shafik@localhost target_dir]$
```

Change to a directory

Directory:

Need execute (x) only allows you to enter/traverse

```
4. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ chmod 100 ~/mydir/
[mina_shafik@localhost ~]$ cd ~/mydir/
[mina_shafik@localhost mydir]$ ls ~/mydir/
ls: cannot open directory '/home/mina_shafik/mydir/': Permission denied
[mina_shafik@localhost mydir]$
```

List a directory content (ls command)

Directory:

r to list file names

x to access/traverse the directory

ls -l also requires x to see file details.

```
4. 192.168.56.101 (root)
[mina_shafik@localhost mydir]$ mkdir ~/list_dir
[mina_shafik@localhost mydir]$ touch ~/list_dir/file1
[mina_shafik@localhost mydir]$ chmod 500 ~/list_dir
[mina_shafik@localhost mydir]$ ls ~/list_dir/
file1
[mina_shafik@localhost mydir]$ chmod 400 ~/list_dir
[mina_shafik@localhost mydir]$ ls ~/list_dir/
ls: cannot access '/home/mina_shafik/list_dir/file1': Permission denied
file1
[mina_shafik@localhost mydir]$
```

View a file content (more/cat command), Modify a file content

File:

r → to read the content

Parent directory:

x → to access the directory to reach the file

```
4. 192.168.56.101 (root)
[mina_shafik@localhost mydir]$ echo "Hello" > ~/myfile
-bash: /home/mina_shafik/myfile: Permission denied
[mina_shafik@localhost mydir]$ ls
ls: cannot open directory '.': Permission denied
[mina_shafik@localhost mydir]$ cd ~
[mina_shafik@localhost ~]$ echo "Hello" > ~/myfile
-bash: /home/mina_shafik/myfile: Permission denied
[mina_shafik@localhost ~]$ ls ~/myfile
/home/mina_shafik/myfile
[mina_shafik@localhost ~]$ chmod ~/myfile
chmod: missing operand after '/home/mina_shafik/myfile'
Try 'chmod --help' for more information.
[mina_shafik@localhost ~]$ chmod 400  ~/myfile
[mina_shafik@localhost ~]$ chmod 500 ~
[mina_shafik@localhost ~]$ cat ~/myfile
[mina_shafik@localhost ~]$ chmod 000 ~/myfile
[mina_shafik@localhost ~]$ cat ~/myfile
cat: /home/mina_shafik/myfile: Permission denied
[mina_shafik@localhost ~]$
```

Modify a file content

File:

w → write to the file

Parent directory:

x → to access the file

```
4. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ chmod 500 ~
[mina_shafik@localhost ~]$ echo "Hello" > ~/modify
-bash: /home/mina_shafik/modify: Permission denied
[mina_shafik@localhost ~]$ chmod 700 ~
[mina_shafik@localhost ~]$ echo "Hello" > ~/modify
[mina_shafik@localhost ~]$ chmod 200 ~/modify
[mina_shafik@localhost ~]$ chmod 500 ~
[mina_shafik@localhost ~]$ echo "World" >> ~/modify
[mina_shafik@localhost ~]$ chmod 400 ~/modify
[mina_shafik@localhost ~]$ echo "Fail" >> ~/modify
-bash: /home/mina_shafik/modify: Permission denied
[mina_shafik@localhost ~]$
```

VIM Editor:

Command to install vim → dnf install vim -y

```
4. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ su - root
Password:
Last login: Sun Mar  8 14:21:15 EET 2026 from 192.168.56.1 on pts/0
[root@localhost ~]# dnf install vim -y
Last metadata expiration check: 3:08:20 ago on Sun 08 Mar 2026 12:20:47 PM EET.
Dependencies resolved.
================================================================================================================================================
 Package                        Architecture                Version                          Repository                                  Size
================================================================================================================================================
Installing:
 vim-enhanced                   x86_64                      2:8.2.2637-23.el9_7              appstream                                  1.7 M
Installing dependencies:
 gpm-libs                       x86_64                      1.20.7-29.el9                    appstream                                   20 k
 vim-common                     x86_64                      2:8.2.2637-23.el9_7              appstream                                  6.6 M
 vim-filesystem                 noarch                      2:8.2.2637-23.el9_7              baseos                                     9.4 k

Transaction Summary
================================================================================================================================================
Install  4 Packages

Total download size: 8.4 M
Installed size: 34 M
Downloading Packages:
(1/4): vim-filesystem-8.2.2637-23.el9_7.noarch.rpm                                          12 kB/s |  9.4 kB     00:00
(2/4): gpm-libs-1.20.7-29.el9.x86_64.rpm                                                    22 kB/s |   20 kB     00:00
(3/4): vim-enhanced-8.2.2637-23.el9_7.x86_64.rpm                                           549 kB/s |  1.7 MB     00:03
(4/4): vim-common-8.2.2637-23.el9_7.x86_64.rpm                                              1.2 MB/s |  6.6 MB     00:05
------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                       1.4 MB/s |  8.4 MB     00:06
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                        1/1
  Installing       : gpm-libs-1.20.7-29.el9.x86_64                                                                                          1/4
  Installing       : vim-filesystem-2:8.2.2637-23.el9_7.noarch                                                                              2/4
  Installing       : vim-common-2:8.2.2637-23.el9_7.x86_64                                                                                  3/4
  Installing       : vim-enhanced-2:8.2.2637-23.el9_7.x86_64                                                                                4/4
  Running scriptlet: vim-enhanced-2:8.2.2637-23.el9_7.x86_64                                                                                4/4
  Verifying        : vim-filesystem-2:8.2.2637-23.el9_7.noarch                                                                              1/4
  Verifying        : gpm-libs-1.20.7-29.el9.x86_64                                                                                          2/4
  Verifying        : vim-common-2:8.2.2637-23.el9_7.x86_64                                                                                  3/4
  Verifying        : vim-enhanced-2:8.2.2637-23.el9_7.x86_64                                                                                4/4

Installed:
  gpm-libs-1.20.7-29.el9.x86_64  vim-common-2:8.2.2637-23.el9_7.x86_64  vim-enhanced-2:8.2.2637-23.el9_7.x86_64  vim-filesystem-2:8.2.2637-23.el9_7.noarch
```

1. Using vi write your CV in the file mycv. Your CV should include your name, age, school, college, experience,...

```
4. 192.168.56.101 (root)
[root@localhost ~]# vi mycv
[root@localhost ~]# vi mycv
[root@localhost ~]# ls -l mycv
-rw-r--r--. 1 root root 243 Mar  8 15:37 mycv
[root@localhost ~]# cat mycv
Name: Mina Shafik
Age: 22
Institute: ITI Benisuef
College: Helwan  University
Experience:
  - Internship at TechCorp, 6 months
  - Volunteer at Open Source Club
Skills:
  - Linux, Bash scripting
  - Python, C++
  - Teamwork and communication
[root@localhost ~]#
```

2. Open mycv file using vi command then: Without using arrows state how to:

a. Move the cursor down one line at time.

j → moves the cursor down one line

b. Move the cursor up one line at time.

k → moves the cursor up one line

c. Search for word age

The cursor will jump to the first occurrence of “age”.

Press n jump to next occurrence.

Press N jump to previous occurrence.

d. Step to line 5 (assuming that you are in line 1 and file is more than 5 lines).

Moves the cursor **directly to line 5**.
(Example: 1G → first line and so on, G alone → last line)

e. Delete the line you are on and line 5.

Delete the line you are currently on: dd

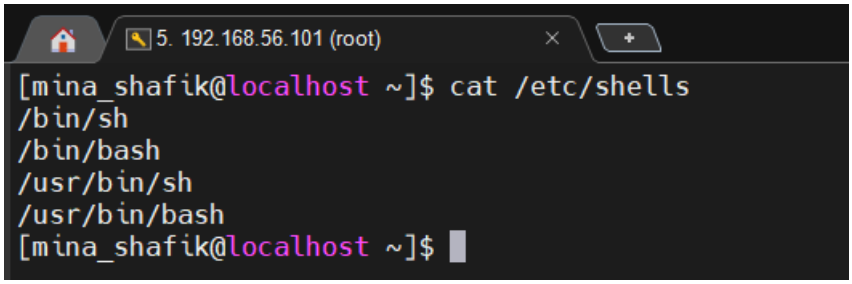
Delete line 5 specifically: Go to line 5: 5G

Delete that line: dd

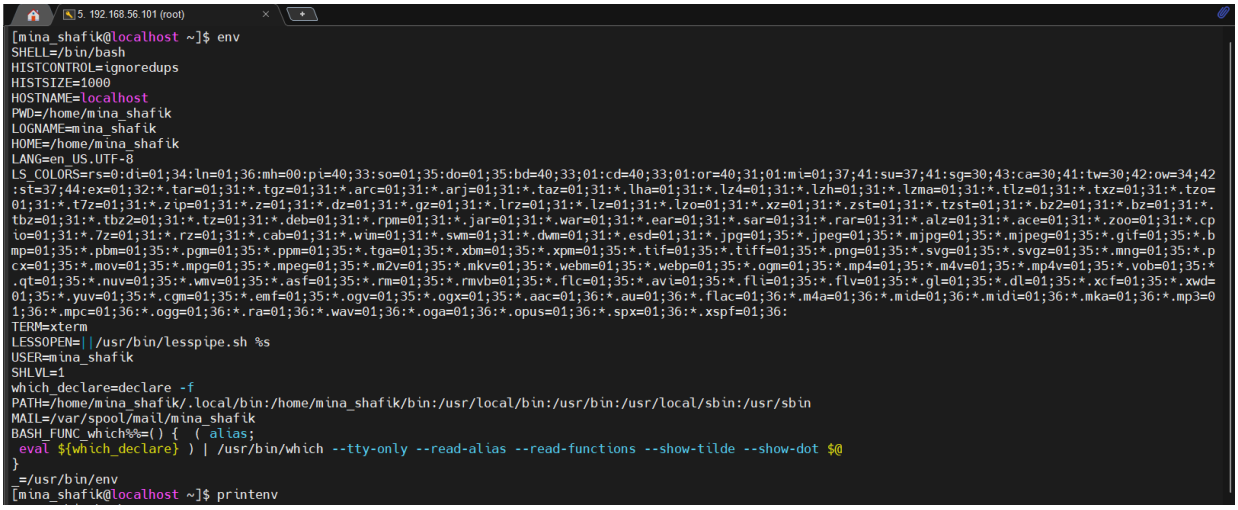
f. How to step to the end of line and change to writing mode in one-step

Press A

3. List the available shells in your system.



4. List the environment variables in your current shell.



5. What are the commands that list the value of a specific variable?

```
[mina_shafik@localhost ~]$ echo $VARIABLE_NAME

[mina_shafik@localhost ~]$ echo $HOME
/home/mina_shafik
[mina_shafik@localhost ~]$ printenv VARIABLE_NAME
[mina_shafik@localhost ~]$ printenv PATH
/home/mina_shafik/.local/bin:/home/mina_shafik/bin:/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin
[mina_shafik@localhost ~]$ env | grep VARIABLE_NAME
[mina_shafik@localhost ~]$ env | grep USER
USER=mina_shafik
[mina_shafik@localhost ~]$
```

6. Execute the following command :

echo \ then press enter

What is the purpose of \ ?

Notice the prompt ">" what is that? and how can you change it from ">" to ":".

```
5. 192.168.56.101 (root)
[mina_shafik@localhost ~]$ echo \
> clear
clear
[mina_shafik@localhost ~]$ $ echo \
> hi this is mina
-bash: $: command not found
[mina_shafik@localhost ~]$ PS2=": "
[mina_shafik@localhost ~]$ echo \
: hello
hello
[mina_shafik@localhost ~]$
```

7. Create a Bash shell alias named ls for the "ls –lta" command

```
# .bashrc

# Source global definitions
if [ -f /etc/bashrc ]; then
        . /etc/bashrc
fi

# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]
then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH

# Uncomment the following line if you don't like systemctl's auto-paging feature:
# export SYSTEMD_PAGER=

# User specific aliases and functions

alias rm='rm -i'
alias cp='cp -i'
alias mv='mv -i'
alias ls='ls -lta'
```